

NAGANO, Japan

WMO: 476100

Lat: 36.6626N

Lon: 138.1920E

Elev: 1378

StdP: 13.98

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.9	22.1	13.1	11.3	28.4	15.4	12.6	28.5	18.3	37.0	15.9	36.2	3.2	270	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.9	91.7	74.8	89.2	74.0	86.6	73.0	76.5	87.5	75.4	85.5	74.3	83.5	7.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.5	131.2	81.0	72.5	126.8	80.3	71.5	122.6	79.7	41.0	87.6	39.9	85.8	38.9	83.7	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.5	15.4	13.7	DB	15.5	96.1	2.0	1.8	14.0	97.3	12.8	98.4	11.7	99.4	10.2	100.6
			WB	14.6	78.5	1.8	1.2	13.2	79.4	12.2	80.1	11.2	80.8	9.8	81.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.1	31.4	33.2	40.5	52.0	62.6	69.7	77.2	79.2	70.9	59.1	46.9	36.7
	DBStd	17.39	4.31	5.22	6.33	7.21	5.53	4.56	4.47	4.39	6.16	5.97	5.76	5.31
	HDD50	1957	577	469	302	62	1	0	0	0	0	4	129	413
	HDD65	4842	1042	889	759	393	110	12	0	0	16	200	544	877
	CDD50	3809	0	0	9	121	392	591	842	905	626	287	35	1
	CDD65	1218	0	0	0	2	36	152	377	440	193	18	0	0
	CDH74	8676	0	0	0	35	315	772	2769	3640	1086	59	0	0
	CDH80	2888	0	0	0	4	50	149	980	1402	296	7	0	0
Wind	WSAvg	5.5	4.5	5.1	6.2	6.7	6.7	5.9	5.2	5.6	5.7	5.5	4.8	4.4
Precipitation	PrecAvg	45.8	3.3	2.7	3.0	2.7	3.1	4.6	5.4	4.6	5.7	4.3	3.1	3.3
	PrecMax	54.9	6.3	4.1	4.7	5.8	7.8	7.1	13.5	8.0	10.2	13.3	5.7	6.1
	PrecMin	29.6	1.6	1.5	1.2	0.8	1.5	2.8	1.3	0.9	1.2	1.1	1.2	1.9
	PrecStd	5.4	1.1	0.7	0.9	1.2	1.5	1.3	2.7	1.8	2.1	2.6	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.9	56.1	67.3	78.8	85.1	87.5	93.9	95.0	90.5	80.0	67.1	57.5
		MCWB	41.3	46.0	50.0	57.4	63.6	69.0	74.8	75.1	73.1	66.9	55.6	49.6
	2%	DB	44.1	49.9	61.6	72.9	80.8	83.9	91.2	92.4	86.9	74.7	62.8	51.8
		MCWB	38.0	41.7	47.5	54.9	61.6	67.6	74.9	75.1	72.0	63.5	53.4	45.2
	5%	DB	41.0	45.5	56.3	68.9	77.6	81.3	88.6	90.2	83.4	71.5	59.7	48.5
		MCWB	35.9	38.7	45.3	53.0	59.8	67.1	74.1	74.8	70.7	61.7	52.3	43.1
	10%	DB	38.6	42.1	52.0	65.0	74.1	78.7	85.7	87.7	80.1	68.3	56.8	45.5
		MCWB	34.3	36.7	43.5	52.0	58.8	66.2	73.2	74.0	69.4	59.8	50.7	40.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.1	48.6	53.3	62.0	67.4	73.5	77.9	78.3	75.0	68.6	58.4	51.3
		MCDB	47.6	54.4	62.4	69.8	77.7	81.7	89.9	90.7	85.8	77.0	62.5	55.1
	2%	WB	39.0	42.5	49.5	57.9	64.6	71.2	76.3	76.9	73.3	65.7	55.7	46.8
		MCDB	43.2	48.1	57.3	68.6	75.7	78.7	87.3	88.1	82.8	72.4	60.5	50.9
	5%	WB	36.6	39.4	46.6	55.5	62.3	69.6	75.1	75.8	72.0	63.4	53.5	43.4
		MCDB	40.1	44.5	55.0	65.3	72.3	77.1	85.2	86.3	80.5	69.5	58.2	47.9
	10%	WB	34.7	37.0	43.9	53.2	60.7	68.0	74.0	74.8	70.7	61.2	51.4	40.9
		MCDB	38.0	41.8	51.4	62.3	70.8	75.7	83.1	84.4	78.4	66.8	56.3	44.9
Mean Daily Temperature Range	5% DB	MDBR	12.4	14.6	17.9	20.8	20.4	16.6	15.4	15.9	14.8	15.7	16.3	13.5
		MCDBR	16.5	20.5	27.0	27.1	26.4	20.9	19.1	19.0	17.9	18.6	19.8	17.9
		MCWBR	11.2	13.1	14.9	12.9	11.1	8.4	7.0	6.5	6.8	8.8	11.5	11.9
	5% WB	MCDBR	15.3	19.3	24.0	22.1	21.1	16.2	17.4	17.5	15.8	14.9	16.5	16.7
	MCWBR	10.9	13.1	14.6	11.9	10.3	7.2	6.8	6.5	6.4	7.9	10.7	11.9	
Clear-Sky Solar Irradiance	taub	0.381	0.421	0.487	0.528	0.547	0.581	0.569	0.569	0.496	0.442	0.398	0.380	
	taud	1.989	1.902	1.835	1.839	1.850	1.836	1.933	1.928	2.078	2.136	2.178	2.094	
	Ebn at Noon	243	249	245	243	241	232	234	230	241	242	239	236	
	Edh at Noon	43	52	62	65	65	66	60	59	48	41	35	36	
All-Sky Solar Radiation	RadAvg	637	892	1197	1488	1680	1551	1494	1512	1209	986	725	568	
	RadStd	44	62	84	119	172	104	217	152	99	91	56	56	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	+0.90	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page